Problem	Penalty	Free	Informed	COLSHADE	L-SHADE Deb	L-SHADE ε-long
lens/original	1.3 ± 0.7	**1.0 ± 1.0**	1.3 ± 1.2	1.7 ± 2.0	2.4 ± 1.8	2.2 ± 2.2
lens/realistic_draft	1.3 ± 0.8	**0.9 ± 0.5**	1.7 ± 1.3	2.2 ± 1.9	3.4 ± 2.8	4.2 ± 2.4
cec/RC12	2.4 ± 1.4	2.8 ± 2.2	**1.7 ± 0.7**	8.5 ± 0.8	15.3 ± 1.4	18.6 ± 3.4
cec/RC13	**0.7 ± 0.4**	**0.7 ± 0.4**	0.8 ± 0.4	1.5 ± 1.4	1.4 ± 1.1	5.4 ± 11.3
cec/RC15	1.3 ± 0.4	**1.1 ± 0.6**	1.4 ± 0.5	3.5 ± 1.6	4.7 ± 1.9	9.8 ± 5.2
cec/RC16	16.4 ± 9.0	**7.9 ± 5.1**	13.3 ± 8.5	32.6 ± 5.2	30.6 ± 2.4	35.5 ± 4.2
cec/RC18	0.8 ± 1.0	0.4 ± 0.9	0.5 ± 1.1	**0.1 ± 0.1**	**0.1 ± 0.1**	**0.1 ± 0.1**
cec/RC21	**0.0 ± 0.0**	**0.0 ± 0.0**	**0.0 ± 0.0**	**0.0 ± 0.0**	**0.0 ± 0.0**	**0.0 ± 0.0**
cec/RC25	11.0 ± 19.2 [2/75 never]	**3.5 ± 1.9**	4.1 ± 3.3	10.1 ± 5.5	9.8 ± 4.9	22.5 ± 7.5
cec/RC26	17.9 ± 14.5 [1/75 never]	26.9 ± 26.8 [4/75 never]	**15.8 ± 21.1 [3/75 never]**	76.1 ± 3.1	61.9 ± 5.1	63.3 ± 5.2
Mean	5.3 ± 2.7	4.5 ± 2.8	**4.1 ± 2.2**	13.6 ± 0.8	13.0 ± 0.9	16.2 ± 1.3

First feasible point, % of the evaluation budget (lower = faster); runs that never become feasible count as 100 %.

Mean ± std over test runs: LLaMEA arm = best algorithm of each of the 5 LLaMEA runs × 15 seeds; baselines = 15 seeds. Mean row: per test run, mean over the problems. Bold: best of the row.